GG-M-00108D (DSA-DM)
June 4, 1973
SUPERSEDING
Int. Fed. Spec.
GG-M-00108c (DSA-DM)
September 22, 1971 and
Interim Revision of
Fed. Spec. GG-M-108a
September 25, 1956

INTERIM FEDERAL SPECIFICATION

MANDRELS, DENTAL HANDPIECE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-M-108a.

1. SCOPE

1.1 Scope. This specification covers mandrels suitable for use with unmounted wheels, disks, polishing cups and brushes.

1.2 Classification. Mandrels covered by this specification shall be of the following classes as specified (see 6.1).

Class AL - Latch, angle handpiece.
Class S - Straight handpiece.
Class AF - Friction grip, angle handpiece.

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metal, Test Methods.

(Activities outside the Federal government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Alanta, Chicago, Kansas City, Mo., Denver, San Francisco, Washington, D. C., Dallas, Los Angeles, and Seattle.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

20675	Cup, Polishing, Dental Handpiece
20676	Brush, Polishing, Dental Handpiece
23065	Mandrels, Dental Handpiece
23290	Mandrel, Dental Handpiece

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 8-69	Tension Testing of Metallic Materials.
E 18-67	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92-72	Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General requirements. The mandrels shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, and function, fit, assembly or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Material. The mandrels including washer shall be fabricated from AISI type 303 or 416 corrosion-resisting steel in conformance with GG-I-526.

3.4 Style, design and dimensions. Style, design, and dimensions of mandrels shall be in accordance with Defense Personnel Support Center (DPSC) Drawings 23065 and 23290 respectively for the applicable class specified. Mandrels shall be suitable for use with either belt or air-driven handpieces. Class AL mandrels shall be suitable for use in latch-type, contra-angle handpieces. Class S mandrels shall be suitable for use with straight handpieces. Class AF mandrels shall be suitable for use with friction-grip handpieces.

3.5 Construction. The mandrels shall consist of a mandrel body, washer and screw. Mandrel body shall be a one-piece construction. Mandrels shall accept and retain securely unmounted wheels and disks, screw-in type polishing cups and brushes in accordance with Defense Personnel Support Center (DPSC) Drawings 20675 and 20676 respectively (see 6.3).

3.6 Corrosion resistance.

3.6.1 Boil test. The mandrels shall show no corrosion when subjected to the boil test as specified in 4.4.2.

3.6.2 Copper sulfate. The mandrels fabricated from AISI type 416 steel shall be tested for resistance to copper sulfate, as specified in 4.4.3.

3.7 Runout. The total indicated runout of the mandrels shall be no greater than 0.002 inch for class AF mandrels, and no greater than 0.003 inch for class AL and S mandrels, when tested as specified in 4.4.6.

3.8 Hardness and tensile strength. The mandrels shall have a minimum Rockwell hardness of 69 on the 15-N scale (Rockwell B85) when tested as specified in 4.4.4; or, as an alternate, a minimum tensile strength of 85,000 pounds per square inch (p.s.i.) when tested as specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.9 Finish. The mandrels shall be smoothly finished and shall have a bright polish. Mandrels shall be free of defects such as cracks, burs and other defects.

3.10 Surface roughness. The surface roughness of class AF mandrel shanks shall be no more than the root-mean-square (RMS) average deviation of 20 microinches in the longitudinal direction of the shank, when determined as specified in 4.4.7.

3.11 Identification marking. Marking with the name or trademark of the manufacturer or contractor on each mandrel is optional.

3.12 Workmanship. Workmanship shall be first class throughout. Mandrels shall be free from defects which detract from their appearance or impair their serviceability, proper functioning and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the sample size, the lot size, the name of the material, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table I. The unit of product for examination purposes shall be one mandrel.

Table I. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product shall be one mandrel. The acceptance number shall be zero for all sample sizes.

Table II. Test Criteria

Characteristic	Requirement	Test procedure	Inspection level
Chemical analysis	3.3	4.4.1	(see 4.2.3)
Boil test	3.6.1	4.4.2	S-2
Copper sulfate	3.6.2	4.4.3	S-2
Hardness	3.8	4.4.4	S-2
Tensile Strength	3.8	4.4.5	S-2
Runout	3.7	4.4.6	S-2
Surface roughness	3.10	4.4.7	S-2

NOTE: It is not necessary to perform both hardness test and tensile strength test. Either test may be performed.

4.2.3 Sampling for raw materials. Sufficient quantity of each lot of metal used in the construction of the mandrels shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.3 Examination. The mandrels shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any mandrel in the sample containing one or more defects shall be rejected, and if the number of defective mandrels in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The mandrels shall be examined for defects including, but not limited to, those listed in Table III.

Table III. Classification of Defects

Categories	Defects
Major A:	
101	Class not as specified.
102	Shank diameter not as specified (all classes).
103	Latch end dimensions not as specified (class AL).
104	Screw binds or does not fit mandrel properly.
105	Screw head not threaded with 64 threads per inch.
106	Screw missing.
Major B:	
151	Latch end (class AL) not cleanly machined.
152	Shank end not smoothly rounded (class AF).
153	Washer missing.
Minor A:	
201	Finish not as specified.
202	Packaging, packing and marking not as specified.
Minor B:	
251	Dimensions (other than 102 and 103) not as specified.

4.4 Test procedures.

4.4.1 Chemical analysis. In the absence of certificates of quality, chemical analysis of the mandrels shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

4.4.2 Boil test. Boil test shall be performed in accordance with the method specified in GG-I-526.

4.4.3 Copper sulfate. Resistance to copper sulfate shall be determined in accordance with the method specified in GG-I-526.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.4 Hardness. The hardness shall be determined in accordance with ASTM Designation E 18-67 or E 92-72.

4.4.5 Tensile strength. The tensile strength shall be determined in accordance with ASTM Designation E 8-69.

4.4.6 Runout. The total indicated runout of the head of the mandrel shall be measured to the nearest 0.0005 inch. The mandrel shall be held in a vee-block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the mandrel and shall be so positioned that the foot contacts the head of the mandrel. The difference between the maximum and minimum readings for one full rotation of the mandrel shall be taken as the total indicated runout.

4.4.7 Surface roughness. The surface roughness of class AF mandrels shall be determined by using a Brush Surfindicator, or other equipment of equal accuracy. The RMS average shall be determined by squaring the height measurements of the peaks and valleys of the surface of the shank, obtaining the averages of the squared values, and then extracting the square root of this average square.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing six mandrels, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit (six mandrels) shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-676, except that commercial colors will be acceptable. The box shall be designed with suitable interior fittings to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall conform to the special requirements as specified in Table II of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements as specified in Table I of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6520-00-117-7316	MANDREL, DENTAL HANDPIECE, Friction Grip Angle Handpiece, Screw Type, 6s.
6520-00-926-8845	MANDREL, DENTAL HANDPIECE, Latch Type Angle Handpiece, Screw Type, 6s: 64 threads per inch.
6520-00-926-8846	MANDREL, DENTAL HANDPIECE, Straight Handpiece, Screw Type, 6s: 64 threads per inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

6.3 The applicable National Stock Number (NSN) for CUP, POLISHING, DENTAL HANDPIECE, Rubber, 24s is NSN 6520-00-982-9377, and for BRUSH, POLISHING, DENTAL HANDPIECE is NSN 6520-00-890-1565 (see 3.5).

6.4 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.5 Limitations. This specification does not cover all types, classes, styles, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1358

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

GG-M-00108D (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.